ChatGPT **Complete Balloon Pop lab**

Completed in the new **BalloonPop folder**.

- Fixed circular click detection, including edge clicks.
- Added normal (+1), bonus (+3), and penalty (−2) balloons.
- Added 3 lives; each missed balloon costs one.
- Added a 30-second round, final score, and full restart.

All **20 automated checks passed**, and both game screens were visually verified.

Double-click **Play.cmd** to play. Nothing was published or pushed to GitHub.

Edited 6 files
+310 -21

outputs/BalloonPop/balloon-pop/game/balloon.py	+19 -9
outputs/BalloonPop/balloon-pop/game/click_detection.py	+3 -2
outputs/BalloonPop/balloon-pop/main.py	+14 -9

Show 3 more files

tell me what the bugs were and how they were cleared

The starter had **one actual bug** and **three missing features**:

- **Click detection:** It compared squared distance with the plain radius, so only clicks near the center worked. I changed the comparison to `distance_squared <= radius * radius`, making the whole circle clickable.
- **Balloon types:** Every balloon awarded the same points. I added distinct colors and scoring: normal **+1**, bonus **+3**, and penalty **−2**.
- **Lives:** Missed balloons disappeared without consequences. I added **3 lives**, deducted one whenever a balloon fully passed the bottom, and ended the game at zero.
- **Timer and restart:** The game ran indefinitely. I added a **30-second countdown**, stopped movement, spawning, and scoring when the round ended, and displayed the final score. Restart clears balloons and resets score, lives, timer, and spawning.

I also made movement independent of frame rate and ensured overlapping balloons pop only one at a time. **All 20 automated checks passed.**